Floating
Ball Valves

FLOATING - Ball Valves

Introduction

The Floating Ball Valves “are **TIV Valves** products” TIV Valves is an Italian manufacturer of quality ball valves striving to be your most valuable partner in the Oil & Gas Industry. TIV Valves is a **Pietro Fiorentini** company.
Based in the northern Italy, TIV Valves was founded in January 2010 to fill a void in high quality engineered ball valves for the Oil&Gas market.

Since then, TIV shipped more than **25,000 valves** in five continents to all main end users and EPC’s companies. TIV can meet simple yet crucial requirements with high quality Italian design, concentrated production lots and short lead times.
TIV provides customized valves to fit a wide range applications. **Severe service** designed valves include corrosive and abrasive fluids, high temperature, cryogenic, underground and any special customer requirement.
The main reference standards are API 6D, API 6A, API 6DSS, ASME B16.34 and ISO 15848.

TIV Valves is able to meet the most severe testing requirements within the Oil&Gas industry in terms of general performance and functionality, high pressure gas testing, high temperature, low temperature, fire safe, fugitive emissions and NDE.
TIV can provide total service and support with its valves. If requested, testing and quality control procedures can be conducted on site.

Classification and Operating Range

Ball valves are cut off devices suitable for use both on natural gas distribution network and for liquid service when high performance on thightness at both high and low differential pressure is required.

The main specifications of these valves are:
- **Steel body** with tail pieces fit for the flanged coupling or with machining for butt welding
- **Soft or metal** seated configuration to ensure proper tightness capability with any kind of process fluid

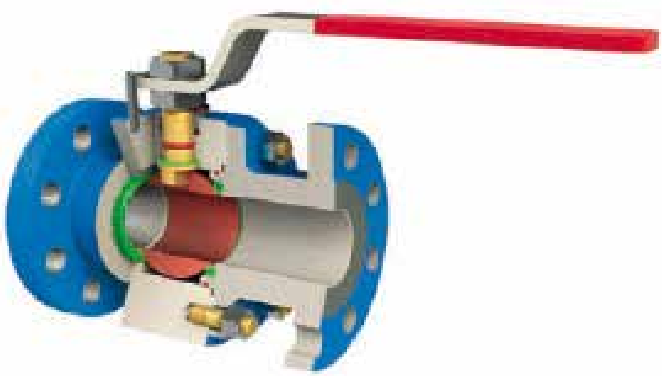


Fig.1 **Floating ball valve**

Product Range

Product Range Floating Ball Valves							
Size	**Class**	**150**	**300**	**600**	**900**	**1500**	**2500**
Inches	**Design**	**SB**	**SB**	**SB**	**SB**	**SB**	**SB**
1,5	FB	■	■	■	■	■	■
	RB	■	■	■	■	■	■
3/4	FB	■	■	■	■	■	■
	RB	■	■	■	■	■	■
1	FB	■	■	■	■	■	■
	RB	■	■	■	■	■	■
1 1/2	FB	■	■	■	■	■	■
	RB	■	■	■	■	■	■
2	FB	■	■	■	■	■	
	RB	■	■	■	■	■	
3	FB	■	■	■	■		
	RB	■	■	■	■		
4	FB	■	■	■			
	RB	■	■	■			
6	FB	■	■				
	RB	■	■				

Tab.1

Features

Design Features

- Anti-blowout Stem
- Antistatic Design
- Fire Safe Design
- Soft Seated or Metal Seated
- Low Fugitive Emission
- Bolted Body
- Cryogenic Extended Bonnet

Severe Applications

- Corrosive
- Erosive
- High Temperature
- Cryogenic
- Lethal service

Materials

- Carbon Steel
- Low Temperature Carbon Steel, Impact Tested
- Ferritic/Austenitic Stainless Steel (Duplex - Super Duplex)
- Austenitic Stainless Steel
- Austenitic Stainless Steel (6Mo)
- Hardened Stainless Steel - Precipitation
- High Strength Low Alloyed Steel
- Titanium
- Exotic Materials

Reference Standards

API Q1	Specification for Quality Programs for the Petroleum, Petrochemical and Natural Gas Industry
ISO TS 29001	Petroleum, petrochemical and natural gas industries - Sector-specific quality management systems Requirements for product and service supply organizations
API 6D	Specification for Pipeline Valves
API 598	Valve Inspection and Testing
API 607	Fire Test for Quarter-turn Valves and Valves Equipped with Nonmetallic Seats
API 6FA	Specification for Fire Test for Valves
API 608	Metal Ball Valves - Flanged, Threaded and Welding Ends
ASME	Boiler and Pressure Vessel Code, Sect. VIII Div.1 & 2
ASME B16.34	Valves Flanged, Threaded and Welding End
ASME B16.10	Face to Face and End-to-End Dimensions of Valves
ASME B16.5	Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24
ASME B16.25	Buttwelding Ends
ASME B1.1	Unified Inch Screw Threads, UN and UNR Thread Form
ASEM B1.20.1	Pipe Threads, General Purpose (Inch)
ASME B1.5 ACME	Screw Threads
ASME B16.20	Metallic Gaskets for Pipe Flanges: Ring-Joint, Spiral-Wound, and Jacketed
ASME B31.3	Process Piping
ASME B31.8	Gas Transmission and Distribution Piping Systems
ASME B36.10	Welded and Seamless Wrought Steel Pipe
BS 6364	Specification for valves for cryogenic service
EN 12266-1	Industrial valves - Testing of metallic valves - Part 1: Pressure tests, test procedures and acceptance criteria - Mandatory requirements
EN 12266-2	Industrial valves - Testing of metallic valves - Part 2: Tests, test procedures and acceptance criteria - Supplementary requirements
EN ISO 17292	Metal ball valves for the petroleum, petrochemical and allied industries
EN 473 / ISO 9712	Non-destructive testing - Qualification and certification of NDT personnel
ISO 10497	Testing of valves - Fire type-testing requirements
ISO 15848	Industrial valves - Measurement, test and qualification procedures for fugitive emissions.
Part 1:	Classification system and qualification procedures for type testing of valves.
Part 2:	Production acceptance test of valves.
ISO 5208	Industrial valves - Pressure testing of metallic valves
ISO 9001	Quality management systems - Requirements

Reference Standard

ISO 5211	Industrial valves - Part-turn actuator attachment
PED (97/23/CE)	Pressure Equipment Directive
Atex (94/9/CE)	Explosives Atmospheres Directive
MSS	
MSS SP-06	Standard Finishes for Contact Faces of Pipe Flanges and Connecting-End Flanges of Valves and Fittings
MSS SP-25	Standard Marking System for Valves, Fittings, Flanges and Unions
MSS-SP-44	Steel Pipeline Flanges
MSS-SP-45	Bypass and Drain Connections
MSS-SP-55	Quality Standard for Steel Castings for Valves, Flanges and Fittings and Other
NACE	
NACE MR0175 - ISO 15156	Petroleum and natural gas industries - Materials for use in H2S-containing environments in oil and gas production - Parts 1, 2, and 3
NACE MR 0103	Materials Resistant to Sulfide Stress Cracking in Corrosive Petroleum Refining Environments
ASTM	
ASTM	Material Specification

Certifications

TIV VALVES is certified according to main international standards and is approved by major Oil&Gas and petrochemical companies and EPC's.

- **ISO 9001**
- **ISO 14001**
- **ISO 18001**
- **API Q1**
- **API 6D**
- **FIRE SAFE**
  API607
  API 6FA
  ISO 10497
- **ATEX**
- **SIL 2**
- **SIL 3**
- **EURASEC**

Technical Features

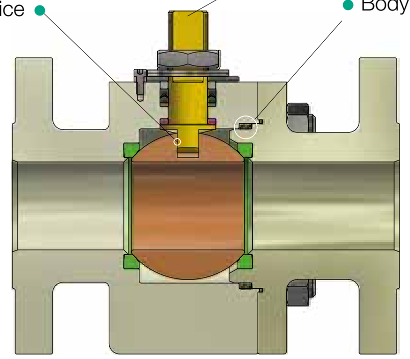


Fig.2 **Floating ball valve** - Typical configuration

Body Joint Sealing

The double sealing action of O – Ring and Graphite, or Lip - seal and Graphite in all static body joint components ensures zero leakage and Fire Safe features. Other special gaskets can be used for special service.

Blow – Out Proof Stem

The stem design is anti blowout. This configuration permits the replacement of the stem seals with the valve in the fully open or closed position.

Anti-Static Device

The Anti-Static Device allows for electrical continuity between valve components.

Extended Bonnet

Extended bonnet ball valves for extreme temperature application are available on request.

Fire Safe

The valves are designed to ensure full functionality in case of fire.

Bi-directional

The valves are designed to block flow in both upstream and downstream directions.

Low Emission

Accurate machining of stem and other sealing surfaces ensure compliance with the most severe pollution-control regulations.

Technical Features

Stem Feature

Valves shall be designed to ensure that the stem does not eject under any internal pressure condition or if the packing gland components and/or valve operator mounting components are removed.
The stem seal integrity is achieved by using special gaskets (PTFE or Graphite Packing, O-Ring, Lip-seal or other design available on request).

NDE's:

- **VT** - Visual Inspection
- **MT** - Magnetic particle test
- **PT** - Liquid penetrant test
- **UT** - Ultrasonic test
- **PMI** - Positive material identification
- **LT** - Fugitive emission test
- **RT** - Radiographic test

Actuators

On request, floating ball valves can be equipped with any actuator type or gearbox. Locking device can be provided as well.

Fig.3 **Basic actuators types**

Floating ball valves - Design

Part List	
Pos.	**Part name**
1	Body
2	Closure
3	Ball
4	Stem
5	Seat
8	Gland Ring
18	Spring Washer
20,21	Seal
30	Gasket
35	Packing
40	Stud Bolt
41,45	Nut
57	Washer
72	Pin
80	Lever
	Tab.2

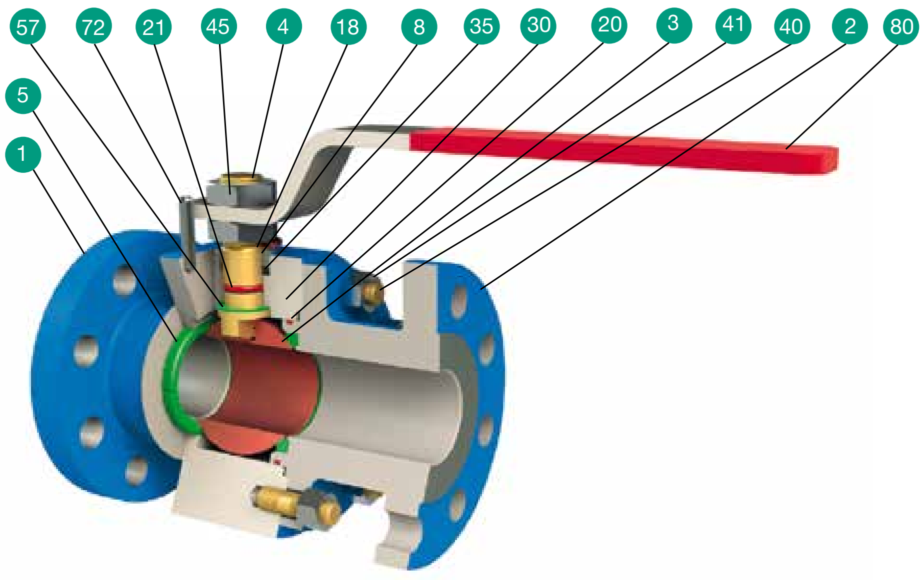
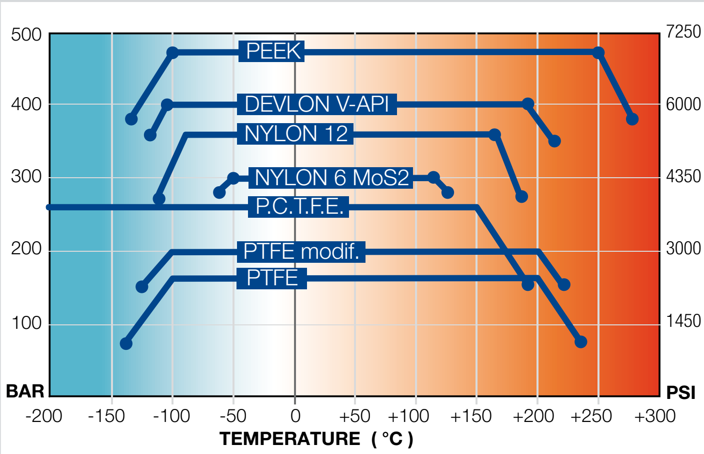


Fig.4 **Floating ball valve** - Main components

Seat Design



Pressure – Temperature rating chart for seat seal materials

Fig.5 **Temperature rating chart**

Standard Material Specification

Material specification					
Pos.	**Part name**	**Carbon steel**	**Low temperature CS**	**Low temperature CS - NACE**	**Stainless steel**
1	**BODY**	ASTM A105	ASTM A 350 LF2 Cl. 1	ASTM A 350 LF2 Cl. 1	ASTM A182 F316
2	**CLOSURE**	ASTM A105	ASTM A 350 LF2 Cl. 1	ASTM A 350 LF2 Cl. 1	ASTM A182 F316
3	**BALL**	ASTM A182 F316	ASTM A182 F316	ASTM A182 F316	ASTM A182 F316
4	**STEM**	ASTM A182 F316	ASTM A182 F316	ASTM A182 F316	ASTM A182 F316
5	**SEAT**	RPTFE	RPTFE	RPTFE	RPTFE
6	**GLAND RING**	ASTM A182 F316	ASTM A182 F316	ASTM A182 F316	ASTM A182 F316
7	**BODY BOLT**	ASTM A193 B7	ASTM A320 L7	ASTM A320 L7M	ASTM A193 B8M
8	**BODY NUT**	ASTM A194 2H	ASTM A194 Gr. 7	ASTM A194 7M	ASTM A194 8M
					Tab.3

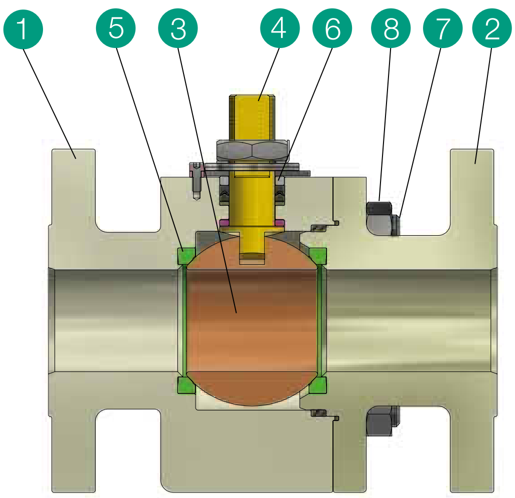
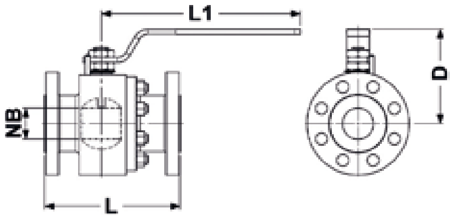


Fig.6 **Floating ball valve** - Part list

Note:
SS, DSS, SDSS or nickel alloys can be requested for pressure containing and pressure controlling components and bolting. Different seat materials available upon request (Devlon-V, PEEK, PTFE, PCTFE, metal to metal) based on the specific application.

Overall dimensions & weight



CLASS 150 Full Bore

SIZE INCH.	NB	L RF	L BW	L1	D	WT. (KG)
1/2	13	108	140	185	90	3
3/4	19	117	152	185	90	4
1	25	127	165	185	95	6
1 1/2	38	165	191	365	125	11
2	49	178	216	365	130	17
3	74	203	283	455	155	28
4	100	229	305	455	170	49
6	150	394	457	800	300	65

Tab.4

CLASS 150 Reduced Bore

SIZE INCH.	NB	L RF	L BW	L1	D	WT. (KG)
3/4 x 1/2	13	117	152	185	90	4
1 x 3/4	19	127	165	185	90	5
1 1/2 x 1	25	165	190	185	95	10
2 x 1 1/2	38	178	216	365	125	15
3 x 2	49	203	283	365	130	25
4 x 3	74	229	305	455	155	42
6 x 4	100	384	457	455	170	60

Tab.5

CLASS 300 Full Bore

SIZE INCH.	NB	L RF	L BW	L1	D	WT. (KG)
1/2	13	140	140	185	90	4
3/4	19	152	152	185	90	6
1	25	165	165	185	95	8
1 1/2	38	190	191	365	125	16
2	49	216	216	365	130	21
3	74	282	283	455	155	40
4	100	305	305	455	170	65
6	150	403	403	800	300	130

Tab.6

CLASS 300 Reduced Bore

SIZE INCH.	NB	L RF	L BW	L1	D	WT. (KG)
3/4 x 1/2	13	140	140	185	90	4
1 x 3/4	19	152	152	185	90	6
1 1/2 x 1	25	165	165	185	95	8
2 x 1 1/2	38	190	191	365	125	16
3 x 2	49	216	216	365	130	21
4 x 3	74	282	283	455	155	40
6 x 4	100	305	305	455	170	65
6	150	403	403	800	300	130

Tab.7

CLASS 600 Full Bore

SIZE INCH.	NB	L RF	L RTJ	L BW	L1	D	WT. (KG)
1/2	13	165	165	165	185	90	7
3/4	19	190	191	191	185	90	9
1	25	216	216	216	185	95	12
1 1/2	38	241	241	241	365	125	19
2	49	292	295	295	365	130	26
3	74	356	359	356	600	155	47

Tab.8

CLASS 600 Reduced Bore

SIZE INCH.	NB	L RF	L RTJ	L BW	L1	D	WT. (KG)
3/4 x 1/2	13	190	191	191	185	90	9
1 x 3/4	19	216	216	216	185	90	13
1 1/2 x 1	25	241	241	241	185	95	18
2 x 1 1/2	38	292	295	292	365	125	24
3 x 2	49	356	359	356	365	130	38
4 x 3	74	432	435	432	600	155	66

Tab.9

CLASS 900 Full Bore

SIZE INCH.	NB	L RF	L RTJ	L BW	L1	D	WT. (KG)
1/2	13	216	216	216	276	95	12
3/4	19	229	229	229	276	95	18
1	25	254	254	254	276	105	24
1 1/2	38	305	305	305	365	130	32
2	49	368	371	368	400	155	43

Tab.10

CLASS 900 Reduced Bore

SIZE INCH.	NB	L RF	L RTJ	L BW	L1	D	WT. (KG)
3/4 x 1/2	13	229	229	229	276	95	14
1 x 3/4	19	254	254	254	276	95	20
1 1/2 x 1	25	305	305	305	276	105	27
2 x 1 1/2	38	368	371	368	365	130	35
3 x 2	49	381	384	381	400	155	49

Tab.11

- Design, weights and dimensions not established by International Standards are subject to change without notice.
- Dimensions in mm

Overall dimensions & weight

CLASS 900 Full Bore

SIZE INCH.	NB	L RF	L RTJ	L BW	L1	D	WT. (KG)
1/2	13	216	216	216	276	95	12
3/4	19	229	229	229	276	95	18
1	25	254	254	254	276	105	24
1 1/2	38	305	305	305	365	130	32
2	49	368	371	368	400	155	43

Tab.12

CLASS 900 Reduced Bore

SIZE INCH.	NB	L RF	L RTJ	L BW	L1	D	WT. (KG)
3/4 x 1/2	13	229	229	229	276	95	14
1 x 3/4	19	254	254	254	276	95	20
1 1/2 x 1	25	305	305	305	276	105	27
2 x 1 1/2	38	368	371	368	365	130	35
3 x 2	49	381	384	381	400	155	49

Tab.13

CLASS 1500 Full Bore

SIZE INCH.	NB	L RF	L RTJ	L BW	L1	D	WT. (KG)
1/2	13	216	216	216	276	95	16
3/4	19	229	229	229	276	95	18
1	25	254	254	254	276	105	26
1 1/2	38	305	305	305	400	130	39

Tab.14

CLASS 1500 Reduced Bore

SIZE INCH.	NB	L RF	L RTJ	L BW	L1	D	WT. (KG)
3/4 x 1/2	13	229	229	229	276	95	18
1 x 3/4	19	254	254	254	276	95	22
1 1/2 x 1	25	305	305	305	276	105	30
2 x 1 1/2	38	368	371	368	400	130	44

Tab.15

CLASS 2500 Full Bore

SIZE INCH.	NB	L RF	L RTJ	L BW	L1	D	WT. (KG)
1/2	13	264	264	264	276	95	18
3/4	19	273	273	273	276	95	25
1	25	308	308	308	276	105	33

Tab.16

CLASS 2500 Reduced Bore

SIZE INCH.	NB	L RF	L RTJ	L BW	L1	D	WT. (KG)
3/4 x 1/2	13	273	273	273	276	95	20
1 x 3/4	19	308	308	308	276	95	28
1 1/2 x 1	25	384	384	384	276	105	36

Tab.17

- Design, weights and dimensions not established by International Standards are subject to change without notice.
- Dimensions in mm

Flow data

Flow Coefficent Cv

SIZE	CLASS					
INCH.	150	300	600	900	1500	2500
1/2	16	14	12	11	11	10
3/4	44	38	34	31	30	29
1	90	78	69	65	63	47
1 1/2	227	211	187	167	163	90
2 x 1 1/2	145	145	140	120	120	100
2	420	420	400	330	330	250
2 1/2	690	690	610	520	510	320
3 x 2	200	200	200	190	180	200
3	1200	1050	1000	910	820	500
4 x 3	600	600	600	590	550	560
4	2200	2100	1850	1800	1700	1100
6 x 4	800	800	790	790	780	745
6	5150	5100	4600	4380	3800	2500
8 x 6	2150	2150	2150	2150	2150	2150
8	9500	9400	9000	8500	7400	5300

Tab.18

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The data are not binding. We reserve the right to make eventual changes without prior notice.

TIV Valves is a Pietro Fiorentini Company, all TIV products are designed and manufactured in Italy in the brand new plant located in Rescaldina near Milan.

TIV VALVES - Italian Plant & Capabilities

25000	10000	3000
m² of total area	m² of of production area	m² offices

CT-s 676-E Mayl 2020